



MAHESH V

Sr Data Engineer

Contact: +1 469-666-1727

LinkedIn: <https://www.linkedin.com/in/mahesh-v-7788vm/>

Email: maheshv0777@gmail.com

Professional Summary:

*Over 6+ years of experience in designing, developing, and implementing **Big Data, Data Warehouse, and Cloud Engineering** solutions using various tools and technologies across domains.*

- Developed **Spark** code using **Scala** and **Spark-SQL/Streaming** for faster testing and processing of data.
- Strong experience in **migrating** other databases to **Snowflake**.
- Good knowledge and experience in **real-time** data streaming technologies like **Spark Streaming** and **Kafka**.
- Configured and created **Kafka Producers** (Python-Client) and **Consumers** for streaming **millions** of events per second on streaming data.
- Proficient in developing **ETL** pipelines that process data in and out of the data warehouse systems using a combination of **Python scripts** and **Snowflake SnowSQL** by executing **SQL queries** against **Snowflake**.
- Experienced in Importing data from **Kafka consumer** into **HBase** using Spark Streaming.
- Extensively involved in designing, reviewing, and optimizing data transformation processes using **Spark** steaming.
- In-Depth knowledge of **Scala** and Experience building **Spark** applications using **Scala**.
- Configured and managed the **DataBricks** clusters for data processing and **data orchestration**.
- Expertise in designing and implementing applications using **Azure Cloud, Azure Data Factory, Azure Data Lake Storage, Azure Synapse, Azure Analytical services, Azure SQL DB, Azure Cosmos NO SQL DB, Azure HD insight Bigdata technologies (Hadoop and Apache Spark), and DataBricks**.
- Experience with the **Hadoop** ecosystem's major components like **Map Reduce, HDFS, YARN, Sqoop, Hive, Pig, Oozie, HBase, Spark, Zookeeper, and Kafka**.
- Good understanding of Hadoop architecture such as **HDFS, Job Tracker, Task Tracker, Name Node, Data Node, and Map Reduce** programming paradigm.
- Experience with **NoSQL** databases that include **HBase, MongoDB, and Cassandra**.
- Extensive experience in **SQL** in query processing, execution, optimization, and performance tuning.
- Expertise in developing scalable **distributed processing** applications utilizing **Spark** and implementing **spark transformations** and **actions** for data processing.
- Experience with setting up ETL jobs on **AWS data pipelines** and **AWS Glue/PySpark**.
- Expertise in **AWS Stack, AWS EC2, S3, IAM, Lambda, Data Pipeline, EMR, SNS, CloudWatch, AWS-Redshift, DMS, and Athena**.
- Expertise with **AWS Lambda function** and **API Gateway** to submit data via API gateway that is accessible via the **Lambda** function.
- Experienced working with different data file formats like **Parquet, Avro, ORC, Textfiles, json**.
- Proficient in monitoring data processing jobs and checking logs in **Cloudera** Manager.

- Developed **PySpark** scripts to process the **data loads** from **upstream systems** and implemented an **in-memory** data computation process to generate the output response for downstream systems effectively.
- Experience in utilizing **Sqoop** to import and export data between **RDBMS** and **HDFS** ecosystem.
- Worked with **Oozie** enabled to run workflow jobs with actions that run **Hive** and **Pig** jobs.
- Hands-on experience in performing analytics on structured data in **Hive** with **Hive queries, views, partitioning, bucketing, and UDFs** using **Hive QL**.
- Experience in **optimizing** and **performance tuning** of Hive queries and **Map Reduce** programs for scalability and faster execution.
- Experience in using **job scheduling** and **monitoring** tools like **Oozie, Airflow, and Zookeeper**.
- Experience with **MS SQL Server integration service (SSIS)**, **SSIS** packages, **T-SQL**, Stored procedures, and Triggers.
- Experience in creating tables, views, triggers, stored procedures, packages, functions, and indexes in **Oracle databases**.
- Planned, tracked, and released program enhancements and bugs in **JIRA** and **ALM**
- Extensive understanding of Software Development Life Cycle (**SDLC**) methodologies such as Waterfall and Agile.
- Experience working with version control systems like **GIT, SVN, Jenkins, and bitbucket** for code development and deployment.

Technical Skills:

Big Data Technologies	HDFS, MapReduce, YARN, Hive, Pig, Spark, Sqoop, Impala, Flume, Oozie, Zookeeper, Kafka
Hadoop Distributions	Cloudera, Hortonworks.
Azure Cloud Platform	Blob Storage, ADLS, Azure Data Factory (ADF), Azure SQL DB, Azure DataBricks, Azure Data Lake, Azure Cosmos DB, Azure Synapse, Azure HDInsight
AWS Cloud Platform	AWS EMR, EC2, S3, Redshift, Athena, Lambda Functions, Step Functions, IAM, API Gateway, AWS Glue.
Programming Languages	Python, Scala, SQL, PL/SQL, T-SQL, HQL, Pig, Shell Scripting, Java
DataBases	Oracle, SQL Server, Postgres, DB2, HBase, Teradata, Cassandra, Snowflake
Operating Systems	Windows, Unix and Linux
ETL Tools	Apache NiFi, Talend, Informatica, Azure Data Factory, Matillion
Version Control Tools	Git, Bit Bucket, SVN
Methodologies	Waterfall, Agile

Professional Experience:

Senior Data Engineer
PepsiCo, Plano, TX

SEP 2021 - Present

Responsibilities:

- Implemented **Spark** scripts using **Scala** and **PySpark** for efficient data processing and utilized **Spark SQL** to access **Hive** tables and perform faster data processing.
- Developed **Spark** scripts using **Scala** shell commands as per project requirements, providing flexibility and customization options for data processing tasks.
- Utilized native **Python** libraries like **NumPy**, **Pandas**, and **Matplotlib** for data processing and analysis. Developed **Python ETL scripts** for handling legacy data loads and data warehousing operations.
- Utilized the **Databricks** platform for developing and executing **Spark**-based data processing loads and Databricks provides a collaborative and scalable environment for working with Spark.
- Implemented **Spark SQL** and **PySpark** scripts for distributed data processing, allowing for parallel processing of large datasets.
- Created pipelines in **Azure Data Factory (ADF)** to extract, transform, and load data from various sources like **Azure SQL**, **Blob storage**, and **Azure SQL Data Warehouse**. ADF facilitated seamless data movement and transformation.
- Developed scalable data ingestion pipelines on **Azure HDInsight Spark cluster** using **Spark SQL**, enabling efficient and reliable data ingestion.
- Integrated **Azure ADF** with **Azure Data Lake** for processing data loads, ensuring seamless data flow and integration between the two services.
- Designed and implemented database solutions using **Azure Synapse Analytics** for efficient data storage and processing, **Azure Synapse Analytics** provides advanced analytics capabilities and integration with other Azure services.
- Used **Azure Blob Storage** as a storage solution for data ingestion and processing, providing scalable and reliable storage for **Bigdata** workloads.
- Leveraged **Azure Functions** for serverless computing and event-driven data processing, enabling automatic scaling and reducing infrastructure management overhead.
- Managed and reviewed **Hadoop log files** for troubleshooting and performance optimization, ensuring smooth operation of the Hadoop ecosystem.
- Designed documentation, data flow patterns, and specifications for near real-time data analytics using **Hadoop** and **HBase**, providing guidelines and standards for effective data processing and analysis.
- Worked with **Hadoop ecosystem technologies** like **Hive**, **Pig**, **Sqoop**, and **MapReduce** to acquire, transform, and publish data in the Hadoop environment.
- Optimized **Hive** queries using techniques such as **map-side joins**, **dynamic partitions**, and **bucketing**, improving query performance and data processing efficiency.
- Designed **Oozie workflows** to run data load jobs into the **Hadoop** system, providing orchestration and scheduling capabilities for data processing tasks.
- Integrated **Kafka** for real-time data streaming and processing, enabling efficient handling of high-volume, real-time data streams.
- Collaborated in an **Agile team** using **Azure DevOps** for project management and version control, ensuring effective collaboration, and streamlined development processes.
- Utilized **Power BI** for data visualization and creating informative dashboards, enabling easy and interactive data analysis and presentation.
- Developed **ETL pipelines** using a combination of **Python** and **Snowflake's SnowSQL** for data warehouse operations, facilitating efficient data integration and analysis.

- Designed data pipelines using **ETL** for effective data ingestion from existing data management platforms to the enterprise **DataLake**, ensuring seamless data flow and integration.
- Created **Grafana dashboards** using structured and unstructured data for monitoring and analytics, providing real-time insights and visualizations.

Environment: Spark, Python, PySpark, Scala, Databricks, Azure ADF, Azure Synapse Analytics, Azure Blob Storage, Azure Function, Hadoop, MapReduce, Kafka, Hive, Pig, Sqoop, Azure DevOps, Power BI

Sr. Data Engineer

SMBC Group, New York, NY

JULY 2019 – AUG 2021

Responsibilities:

- Developed Spark applications using **Scala** and **Python** languages with end-to-end code coverage.
- Utilized **Spark** for faster processing of data and implemented **Spark** scripts in **Python** for data aggregation and validation on **AWS EMR**.
- Stored streaming data to **HDFS** using **Scala** and implemented Spark for faster data processing.
- Worked with Spark **Python libraries** to manipulate data using broadcast joins and Sort merge joins.
- Developed **PySpark programs**, created data frames, and performed transformations.
- Developed **Spark** scripts and **PySpark** programs in **Python** for data processing and aggregation.
- Created **AWS Lambda** functions using Python for deployment management.
- Utilized **Python** for automation of resulting scripts and workflow using **Apache Airflow** and shell scripting.
- Used **Python** to implement serverless architecture with **API Gateway**, **Lambda**, and **DynamoDB**.
- Developed **Spark** applications using **Scala**, leveraging its functional programming capabilities.
- Used **Scala** to store streaming data to **HDFS** and implemented **Spark** for faster processing of data.
- Collaborated with **Databricks** for developing and deploying **Spark** applications.
- Leveraged **Databricks notebooks** for interactive development and data exploration.
- Utilized **Snowflake** for cloud-based data warehousing solutions and performed ETL operations.
- Developed Business Intelligence solutions in Data Warehousing using **Snowflake**.
- Implemented a serverless architecture using AWS services such as **API Gateway**, **Lambda**, and **DynamoDB**.
- Deployed AWS Lambda code from **Amazon S3** buckets.
- Configured **AWS Glue** catalog crawlers to retrieve data from S3 and perform SQL query operations.
- Developed **PySpark** programs and tested their performance on **MR** workloads on **AWS EMR**.
- Collected logs from source systems into **HDFS** using **Kafka** and performed analytics on the data.
- Configured **Kafka** topics and created producers to ingest streaming data.
- Configured consumer groups to process the streaming data events.
- Designed and configured a **Kafka cluster** capable of accommodating heavy throughput of 1 million messages per second.
- Developed multiple **MapReduce** jobs in **Java** for data cleaning and processing.
- Performed **Sqoop** incremental imports on a daily basis using **Oozie**.
- Stored streaming data to **HDFS** using **Scala**.
- Transformed and analyzed data using **Pyspark**, **Hive**, and **Pig** based on **ETL** mappings.
- Utilized **Sqoop** for importing and exporting data between **Hadoop** and relational databases.
- Worked with **MS SQL Server** for database management and integration with data processing pipelines.
- Utilized **GitHub** for version control and collaboration on code repositories.

- Designed, investigated, and implemented public-facing websites on **Amazon Web Services** and integrated them with other application infrastructure.
- Utilized **Tableau** for data visualization and reporting purposes.

Environment: Spark, Python, Scala, Databricks, Snowflake, AWS Glue, lambda, S3, EMR, Kafka, Hadoop, MapReduce, HDFS, Hive, Pig, Sqoop, MS SQL Server, GitHub, Tableau.

Hadoop Developer
Hyperlink Info Systems, India

MAY 2017 – MAY 2019

Responsibilities:

- Worked on large-scale **Hadoop YARN cluster** for distributed data processing and analysis using **Connectors, Hive, Pig, Sqoop, Spark core, Spark SQL, and NoSQL databases**.
- Responsible for developing prototypes of the selected solutions and implementing complex **Bigdata** projects with a focus on collecting, parsing, managing, analyzing, and visualizing large sets of data using multiple platforms.
- Practical exposure and good knowledge in implementing batch and real-time analytics of larger data sets processing using **Pyspark**.
- Develop utilities using Scala and python to minimize the number of lines of code in daily developments.
- Developed spark applications in python (**PySpark**) on the distributed environment to load a huge number of **CSV** files with different schema into **Hive ORC tables**.
- Implement One-time Data Migration of **Multistate level data** from **SQL server** to **Snowflake** by using **Python** and **SnowSql**.
- Implemented **Sqoop** to extract and load data into a **DataLake** environment (**MS Azure**), which was then accessible by business users.
- Responsible for estimating the **cluster size, monitoring, and troubleshooting the Hadoop Cluster**.
- Configured **big data workflows** to run on the top of Hadoop using **Oozie** and these workflows comprise heterogeneous jobs like **Pig, Hive, and Sqoop** Cluster coordination services through **Zookeeper**.
- Created **Pig scripts** to transform the **HDFS** data and loaded the data into **Hive** external table.
- Involved in working with Bigdata **Testing analysis** and **Hadoop** and **Hive** as well.
- Experienced in running **Hadoop stream jobs** to process **terabytes** of **xml group** information with the help of **Map Reduce programs**.
- Creating indexes and tuning the **SQL** queries in **Hive**.
- Migrated computational code in **HQL** to **PySpark**.
- Involved in importing data from **Teradata** into **Hadoop** Ecosystem.
- Planned, tracked, and released program enhancements and bugs in **JIRA** and **ALM**.
- Experience in using **HBase** as the backend database for application development.

Environment: Java, Python, HDFS, MapReduce, Hive, YARN, PIG, Shell Scripting, Scala, Spark, Sqoop, Teradata, Git, HBase.

Responsibilities:

- Experience working in data warehousing environment developed **schemas** for data storage and data processing.
- Designed **Snowflake** and **Star schema OLAP** systems for processing analytical queries on large database systems.
- Extracted data from various heterogeneous sources like **Oracle, Flat Files**, and Excel files and performed complex joiner, Expression, Aggregate, Lookup, Stored procedure, Filter, Router transformation, Update strategy transformations to load data into the target systems and created Sessions, Tasks, Workflows, and Worklets using Workflow manager.
- Created Data marts and connected them to **Tableau** to develop interactive **KPI dashboards**.
- Identify and document limitations in data quality that jeopardize the ability of internal and external data analysts and wrote standard **SQL Queries** to perform data validation and created excel summary reports (**Pivot tables** and **Charts**) as well as gathered analytical data to develop functional requirements using **data modelling**.
- Created complex **SQL queries** and scripts to extract and aggregate data to validate the accuracy of the data and Business requirements gathering and translating them into clear and concise specifications and queries.
- Used **Pandas API** for performing **exploratory data analysis** on **product data** to understand the key insights on **customer usage patterns** and effective areas of improvement.
- Performed data analysis using **google APIs** and created visualizations such as **pie charts, and waterfall charts** and displayed them in the **web application**.
- Worked on sort & filters of **tableau** like **Basic Sorting, basic filters, quick filters, context filters, condition filters, top filters, and filter operations**.
- Worked on **snowflake** multi-cluster warehouses.

Environment: Python, Tableau, SQL, Excel, Data warehousing, Star Schema, Snowflake Schema.